EarthScape Setup Guide

Offline Windows setup

1. Extract the entire ZIP to a normal folder. Do not start the program inside the ZIP preview.

2. Open 05_Project, then EarthScape. Double-click START_EARTHSCAPE.bat.

3. The bundled Python runtime starts the application. No Python installation or internet connection is required for this mode.

4. Open http://127.0.0.1:8080 if your browser does not open automatically. Keep the application window open.

5. Sign in using one of the demo accounts below. Press Ctrl+C in the application window to stop.

Administrator: admin / EarthDemo!2026

Analyst: analyst / AnalystDemo!2026

Viewer: viewer / ViewerDemo!2026

Full Hadoop setup

Install and start Docker Desktop. Open 05_Project/EarthScape and run START_FULL_STACK.bat. The first build downloads dependencies and requires internet.

Open http://localhost:8081. The supplied deployment starts the Hadoop lab with HDFS, YARN and two DataNodes. Use STOP_FULL_STACK.bat to stop it.

Local mode uses local mapper/reducer execution. Use full-stack mode to demonstrate distributed Hadoop processing. Both modes use the same website workflows.

Linux or macOS: install Python and the pinned requirements, then run bash start.sh. For Hadoop, use docker compose up --build -d. The bundled Windows runtime only runs on Windows.

Demonstration sequence

Sign in as admin and inspect Overview. Filter a station and open Models.

Import data/sample.csv and inspect duplicate handling. Submit a new sensor reading above 40 degrees Celsius and review its alert.

Archive data/sample_satellite.png, then run a processing job and inspect the station/month output.

Create and resolve a support ticket, create a backup, then sign in as viewer to inspect restricted access.

Watch 04_Demonstration/Website_Demonstration.mp4 for the included walkthrough.

Troubleshooting and handover

If startup fails, read 05_Project/EarthScape/runtime/launcher.log. Ensure the ZIP is fully extracted and the portable folder remains with the application.

If the browser does not open, enter http://127.0.0.1:8080 manually. If port 8080 belongs to another program, stop that program or configure a different PORT before launching app.py.

For queued Hadoop jobs, check Docker container logs and pipeline status. Full-stack operation requires working Docker installation and sufficient local resources.

The application and 18 unit tests plus 10 HTTP integration checks passed on Linux during this package review. The Windows runtime and Docker Desktop launch were not executed here. Included browser/Hadoop records are retained evidence from the supplied project.

Seed observations are synthetic. Forecasts are educational baselines. Demo accounts and encryption keys are supplied for local evaluation.

Before submission, complete 06_Submission/Submission_Details.docx with genuine names, enrollment IDs, batch, group and centre. Fill administrative forms with actual information. Rename the ZIP with batch, group and project title. The supplied process permits one final submission per group.